

# Axel

## Implementation Plan and PRD Review

Based on: Axel PRD (Draft / Scaffold stage, last updated 2026-09-15)

### 1. Executive summary

The PRD's core idea is sound: keep LLMs in the creative layers (idea generation, signal weighing) and keep everything from the risk gate downward deterministic and auditable. Paper-first rollout, withdrawal-disabled keys, and a backtest gate before any strategy is trusted are all the right instincts. This plan turns that into a build order.

The main recommendation is to **build vertically, not horizontally**: ship one section (Stocks on Alpaca paper) end to end, from data to dashboard, before adding Crypto, Options, or Bets. Build the deterministic core **before** any agent, so the safety layer exists before anything can send an order to it.

Three things need fixing in the PRD before coding starts (details in section 2):

- **Axel is a "final trade gate" and an LLM hybrid.** That contradicts the core design principle. The last gate must be deterministic.
- **Risk numbers are ambiguous or inconsistent.** "Kelly capped at 25%" and "5% max position" collide, and four enforcers (Loss Manager, Risk Engine, Profit Optimizer, Position Sizer) do not exist in the architecture.
- **Backtesting LLM agents is not the same as backtesting strategies.** LLMs have seen the historical period in training, so their backtested "judgment" is contaminated. The backtester should validate deterministic rules; the LLM layer must be validated by forward paper trading.

### 2. PRD review

#### 2.1 What is strong

- Hard separation between LLM layers and the deterministic Risk Agent / Execution Bot.
- Signals from analysts, not trade decisions. This keeps the LLM output as data that can be scored.
- Expectancy, Sharpe, and max drawdown as validation metrics; the explicit warning that 4R+ means overfitting or leakage.
- Telegram signals log-only and copy-trading advisory-only; withdrawal-disabled, trade-only API keys.
- Kill switch at 10% total drawdown with manual resume; a human-approval gate for early live trading.

#### 2.2 Issues to resolve before building

| # | Issue                                                            | Why it matters                                                                                              | Recommended fix                                                                                                                                              |
|---|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Axel is "LLM + deterministic hybrid" and the "final trade gate". | An LLM in the last gate breaks the core principle.                                                          | Axel's LLM only emits typed <i>proposals</i> (allocation shifts, trade approvals). A deterministic gate validates, clamps, and can reject every one of them. |
| 2 | "Kelly capped at 25%" vs "max position 5%".                      | Ambiguous: 25% of Kelly, or 25% of capital? If the latter, it always loses to the 5% cap.                   | Define as <b>fractional Kelly</b> = $0.25 \times f^*$ , then <b>min(result, 5% cap)</b> . Log which limit bound.                                             |
| 3 | Position cap is a notional cap, not a risk cap.                  | 5% notional with a wide stop can still lose far more than intended; 3% daily stop is very loose against it. | Add a per-trade <b>risk-at-stop cap</b> (suggest 0.5-1% of section capital). Size = min(Kelly, 5% notional, risk cap / stop distance).                       |

| #  | Issue                                                                                                                | Why it matters                                                                                 | Recommended fix                                                                                                                                                                                                                 |
|----|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | Loss Manager, Risk Engine, Profit Optimizer, Position Sizer are named in the risk table but not in the architecture. | Unowned enforcement means gaps, or duplicated logic.                                           | Make them modules of a single <b>risk package</b> (section 5). All deterministic, all unit-tested.                                                                                                                              |
| 5  | Kill-switch semantics are undefined.                                                                                 | "Pause" is not enough: open options positions and 24/7 crypto keep moving.                     | Define drawdown (peak-to-trough, mark-to-market vs high-water mark), and the action per section: cancel open orders, block new ones, and flatten or hold by explicit policy. Run it as an <b>independent watchdog process</b> . |
| 6  | Panel confidence threshold 0.3.                                                                                      | Verbalized LLM confidence is uncalibrated; 0.3 is a low bar and the number may be meaningless. | Treat confidence as a feature. Track calibration (Brier score, reliability curve) on paper data and set the threshold from that, not a guess.                                                                                   |
| 7  | Backtest requirements lack pass/fail criteria and costs.                                                             | "Backtested and scored" without thresholds lets anything through.                              | Fix thresholds up front (section 8), require fees + slippage, point-in-time data, walk-forward splits, and a multiple-testing adjustment.                                                                                       |
| 8  | Inbound email / news / social feed LLM agents.                                                                       | Prompt injection: a crafted email or headline could steer an analyst.                          | Treat all ingested text as untrusted. Analysts have no side-effect tools, must return schema-validated JSON, and never see credentials.                                                                                         |
| 9  | No human-approval component in the architecture.                                                                     | The non-goals require it, but nothing implements it.                                           | Add an <b>Approval service</b> : in live mode, proposals wait for email/dashboard approval and expire if not approved.                                                                                                          |
| 10 | Missing operational requirements.                                                                                    | 24/7 systems fail in boring ways: stale data, duplicate orders, broker/state drift.            | Add: idempotent orders, broker reconciliation, data-freshness checks, LLM budget cap, heartbeats/alerts, immutable decision log, market calendars, corporate actions.                                                           |
| 11 | Scope: four sections at once.                                                                                        | Each has different data, brokers, math, and legal risk. Parallel work stalls everything.       | Stocks first. Crypto second. Options third. Bets last, gated on legal review per venue.                                                                                                                                         |
| 12 | Venue availability and legal status.                                                                                 | Broker/exchange onboarding, KYC, and prediction-market access differ by country of residence.  | Before writing venue code, confirm you can legally open and fund accounts there. Paper trading needs no such check, which is another reason to start there.                                                                     |

### 2.3 Suggested PRD amendments (summary)

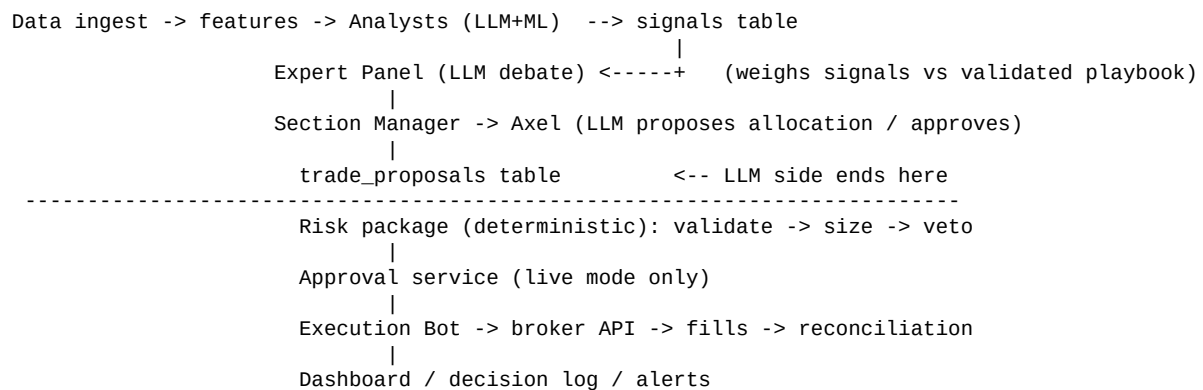
- Reword Axel's row: "LLM proposes; deterministic gate disposes."
- Replace the risk table's "Enforced by" names with modules from the risk package; add per-trade risk cap, gross exposure cap per section, and no-leverage / defined-risk-only rules for Crypto and Options.
- Define kill-switch trigger, measurement, and per-section actions.
- Add an Approval service and an Observability section (alerts, heartbeats, LLM cost budget).
- State that the backtester validates deterministic strategy rules; LLM agent quality is validated by forward paper trading and calibration tracking.
- Add success criteria per phase (section 7) so "done" is testable.

### 3. Target architecture

Three processes with different trust levels. The boundary between them is a database table plus typed schemas, never a direct function call from LLM code into execution code.

| Process       | Trust                   | Contains                                                                                                    | Rules                                                                                                                             |
|---------------|-------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Agent service | Untrusted / creative    | Analysts, Expert Panel, Section Manager, Axel (LLM part), chat interface                                    | Reads data and signals. Writes only <i>signals</i> and <i>proposals</i> to Postgres. No broker credentials. LLM budget guard.     |
| Trading core  | Trusted / deterministic | Risk package (rules, sizer, loss manager, allocator clamp), Execution Bot, reconciliation, approval service | No LLM imports (enforced by a lint/CI check). Consumes proposals, emits risk decisions and orders. Holds trade-only keys.         |
| Watchdog      | Trusted / independent   | Kill switch, heartbeat monitor, broker-vs-DB reconciliation checker                                         | Separate process, separate DB role and broker read access. Can cancel orders and set a global HALT flag that the core must honor. |

#### Data flow for one decision



Every arrow across the dotted line is a typed, versioned record. That gives you a full audit trail and lets you replay any day offline.

### 4. Repository layout and core contracts

```

axel/
  core/          config, clock (injectable time), types (pydantic), ids
  risk/          rules.py sizer.py loss_manager.py allocator.py killswitch.py
  execution/     broker_base.py alpaca.py order_fsm.py reconcile.py approval.py
  data/          ingest/ features/ ml/backtester.py ml/validation.py
  agents/        analysts/ panel.py section_manager.py axel.py llm_budget.py
  sections/      stocks/{config.yaml, playbook/} crypto/ options/ bets/
  comms/         email_in.py email_out.py chat.py
  dashboard/     (Grafana provisioning first, custom UI later)
  infra/         docker-compose.yml migrations/ .env.example
  tests/         unit/ property/ replay/ integration/

```

Contracts (pydantic models, versioned). These are the only things that cross the LLM / deterministic boundary:

```

Signal      {id, ts, as_of, section, symbol, analyst, direction, strength, horizon, features_ref}
Proposal    {id, ts, section, symbol, side, entry, stop, target, strategy_id,
             panel_confidence, rationale_ref, mode: paper|live}
RiskDecision {proposal_id, approved, approved_qty, binding_limit, reasons[]}
Order       {client_order_id (idempotent), proposal_id, symbol, side, qty, type, state}
AllocationProposal {from_section, to_section, pct_of_total} # clamped to 5% per cycle

```

## 5. Risk package specification (clarified)

All rules are pure functions of (proposal, account state, config). No I/O inside rule logic, so they can be unit- and property-tested exhaustively.

| Control                                 | Value (from PRD)            | Clarification / addition                                                                                                                                                            |
|-----------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Max position size                       | 5% of section capital       | Notional cap. Add <b>risk-at-stop cap</b> (suggest 0.5-1% of section capital).                                                                                                      |
| Position sizing                         | Kelly, capped 25%           | size = $\min(0.25 \times \text{Kelly } f^*, 5\% \text{ notional, risk cap} / \text{stop distance})$ . Log the binding limit. If edge estimate $\leq 0$ , size is zero.              |
| Section daily stop-loss                 | 3% of section capital       | Measured on realized + unrealized P&L; since section day start (exchange-local for Stocks, UTC for Crypto). On breach: block new entries, cancel working orders, flatten by policy. |
| Max reallocation                        | 5% of total capital / cycle | Deterministic clamp on any Axel AllocationProposal; reject the excess, do not scale silently.                                                                                       |
| Min panel confidence                    | 0.3                         | Placeholder. Re-derive from calibration data collected in paper mode.                                                                                                               |
| Kill switch                             | 10% total drawdown          | Peak-to-trough on total marked-to-market equity vs high-water mark. Watchdog process. Sets HALT; resume only via manual command with a written reason logged.                       |
| <b>New:</b> gross exposure              | -                           | Per-section and total cap on gross exposure; no leverage in Crypto; Options limited to defined-risk structures until proven.                                                        |
| <b>New:</b> correlation / concentration | -                           | Cap on one symbol/sector/underlying across sections (Axel sees the total book).                                                                                                     |
| <b>New:</b> data and order hygiene      | -                           | Reject on stale quotes, market closed, unusual spread, order rate above limit, or a duplicate client_order_id.                                                                      |

### Kelly reference

For a strategy with win probability  $p$ , loss probability  $q = 1 - p$ , and average win/loss ratio  $b$ :  $f^* = (b \times p - q) / b$ . Use  $0.25 \times f^*$ , then apply the caps above. Estimates of  $p$  and  $b$  come from the out-of-sample backtest and paper history, with a haircut (for example, use the lower bound of a confidence interval), because overestimated edge makes Kelly dangerous.

**Test requirement:** property-based tests (Hypothesis) assert that for any random proposal and account state, no approved order can exceed any configured limit. This is the single most valuable test suite in the project.

## 6. Data model (PostgreSQL + TimescaleDB)

| Table                              | Type                 | Purpose                                                                                    |
|------------------------------------|----------------------|--------------------------------------------------------------------------------------------|
| instruments, calendars             | regular              | Symbols, venues, trading hours, holidays, corporate actions.                               |
| bars, quotes, funding_rates        | hypertable           | Market data, stored with both event time and ingestion time (point-in-time correctness).   |
| news_items, filings                | hypertable           | Raw ingested text with source, trust flag, and hash for dedupe.                            |
| signals                            | hypertable           | Analyst outputs: schema-validated, with model and prompt version.                          |
| panel_decisions, trade_proposals   | regular              | Panel rationale, confidence, and the proposal handed to the core.                          |
| risk_decisions                     | regular              | Every approval and veto with the binding limit and reasons.                                |
| orders, fills, positions_snapshots | regular / hypertable | Order state machine history, executions, periodic position snapshots.                      |
| equity_snapshots                   | hypertable           | Per-section and total equity; feeds drawdown and dashboard.                                |
| strategy_registry                  | regular              | Status: candidate / validated / paper / live; link to backtest report and parameters hash. |

| Table                                   | Type        | Purpose                                                                             |
|-----------------------------------------|-------------|-------------------------------------------------------------------------------------|
| agent_runs                              | regular     | Model, prompt version, tokens, cost, latency, input/output refs for every LLM call. |
| killswitch_events, approvals, audit_log | append-only | No UPDATE/DELETE grants for the application role.                                   |

## 7. Phased build plan

Estimates assume one developer working part-time and are ranges, not commitments. Each phase has an exit test; do not start the next until it passes.

| Phase                          | Deliverables                                                                                                                                                                                                                                                                                                                                                                                 | Exit criteria                                                                                                                                                                 | Est.    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>0. Foundations</b>          | Repo, Docker Compose (Postgres+TimescaleDB), migrations, typed config, structured logging, injectable clock, CI (lint, types, tests), secrets via .env with .env.example, CI check that risk/ and execution/ import no LLM libraries.                                                                                                                                                        | Fresh clone -> one command -> DB up, tests green.                                                                                                                             | 1 wk    |
| <b>1. Deterministic core</b>   | Contracts; risk rules, sizer, loss manager, allocator clamp; order state machine; Alpaca paper broker adapter; idempotent orders; reconciliation job; watchdog + kill switch; email alert on HALT. Drive it with a hand-written dummy proposal script - no LLM yet.                                                                                                                          | Property tests pass. Kill-switch drill: simulate a 10% drawdown, confirm HALT, cancels, alert, and blocked resume. Paper orders round-trip with matching DB and broker state. | 3-4 wks |
| <b>2. Data + backtester</b>    | Ingest Alpaca/Polygon bars, FRED, EDGAR, one news source, all point-in-time. Backtest engine with fees, slippage, market calendar, walk-forward splits. Validation report (expectancy in R, Sharpe, max DD, trade count, parameter sensitivity, multiple-testing adjustment). Two or three simple baseline strategies (for example, trend-following and mean-reversion) as playbook entries. | Backtester reproduces a known result on a textbook strategy. A deliberately overfit strategy is <i>rejected</i> by the validation gate.                                       | 3-4 wks |
| <b>3. Agent layer (Stocks)</b> | Analysts (technical, fundamental, sentiment, macro) emitting schema-validated signals; Expert Panel weighing signals against the validated playbook; Section Manager; Axel proposal logic; LLM budget guard; injection-resistant ingestion. Event-driven triggers, not constant polling.                                                                                                     | End-to-end paper run for a week with zero schema violations, every proposal traceable in the decision log, LLM spend under budget.                                            | 3-4 wks |
| <b>4. Dashboard + comms</b>    | Grafana on TimescaleDB first: P&L, positions, drawdown, exposure, decision log. Email reports/alerts, approval flow, chat interface (read-only at first). Custom UI only if Grafana falls short.                                                                                                                                                                                             | You can answer "why did it take that trade?" from the dashboard alone.                                                                                                        | 2 wks   |
| <b>5. Paper soak (Stocks)</b>  | Run continuously in paper mode. Track calibration, slippage vs backtest assumptions, incident log. Tune thresholds only with written rationale.                                                                                                                                                                                                                                              | Live-promotion gate (section 8) met.                                                                                                                                          | 8+ wks  |
| <b>6. Tiny live (Stocks)</b>   | Small, loss-tolerable capital; manual approval on every trade at first; relax approval gradually.                                                                                                                                                                                                                                                                                            | Live results within tolerance of paper; no risk incidents.                                                                                                                    | ongoing |
| <b>7. Crypto</b>               | Exchange adapter (Binance/Kraken), on-chain and funding-rate analysts, UTC day boundaries, no leverage. Reuse core unchanged.                                                                                                                                                                                                                                                                | Repeat phase 5 exit for Crypto.                                                                                                                                               | 4-6 wks |
| <b>8. Options</b>              | IV data source, Greeks, defined-risk structures only, assignment/expiry handling.                                                                                                                                                                                                                                                                                                            | Pricing model matches vendor Greeks within tolerance; expiry drills pass.                                                                                                     | 6-8 wks |
| <b>9. Bets / Predictions</b>   | Only after written legal review per venue. Calibration-first: measure probability calibration before any staking logic. Kalshi first if permitted.                                                                                                                                                                                                                                           | Legal sign-off recorded; calibration score beats market-implied baseline in paper.                                                                                            | TBD     |

## 8. Validation and promotion gates

The values below are suggested defaults. Fix them **before** looking at any results, and record any later change with a reason. Thresholds chosen after seeing the numbers are a form of overfitting.

### 8.1 Strategy admission to the playbook

| Test                | Suggested default                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Data                | Real historical data only, point-in-time, survivorship-bias free where possible; costs and slippage included.                 |
| Split               | Walk-forward with untouched final out-of-sample block; never tune on it.                                                      |
| Sample size         | At least 100 out-of-sample trades (more for low-frequency strategies).                                                        |
| Expectancy (net, R) | Between roughly 0.1R and 0.5R per trade. Above about 1R triggers a leakage audit, per the PRD.                                |
| Risk-adjusted       | Out-of-sample Sharpe above a threshold you set in advance (for example 0.7 net); max drawdown within the section's tolerance. |
| Robustness          | Small parameter perturbations (+/- 20%) do not flip the sign of expectancy; results hold across at least two market regimes.  |
| Multiple testing    | Track how many variants were tried; apply a deflated-Sharpe or similar adjustment.                                            |

Expectancy (R) = (Win% x Avg Win in R) - (Loss% x Avg Loss in R). Compute it **net of costs**, with R defined by the stop distance recorded at entry.

### 8.2 Validating the LLM layer

- A model trained on data that includes the backtest period can "recall" what happened. Backtests of LLM judgment are therefore unreliable. Use LLMs to *rank and weigh* pre-validated deterministic strategies, and validate that layer forward in paper mode.
- Ablation: run a rules-only baseline in parallel. If the LLM panel does not beat it after costs over the soak period, simplify. This is a real possibility and a fine outcome.
- Track calibration of panel confidence (Brier score) and disagreement between analysts.

### 8.3 Paper-to-live gate (per section)

- At least 8 weeks and 200 paper trades (or an agreed equivalent for low-frequency sections).
- Paper expectancy and slippage within tolerance of backtest assumptions.
- Zero unresolved reconciliation breaks; zero risk-rule bypass incidents; kill-switch drill passed in the last 30 days.
- Recovery drill passed: kill the process mid-order, restart, and state reconciles correctly.
- Explicit manual sign-off, logged. Start with small capital and per-trade human approval.

## 9. Security, cost, and operations

| Area            | Requirement                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Secrets         | Trade-only, withdrawal-disabled keys; one key per section; .env locally, a secrets manager on the server; keys never exposed to the agent service; rotate on any suspicion.                                       |
| Untrusted input | Email, news, social, Telegram are data, not instructions. Wrap in delimiters, strip markup, validate outputs against schemas, allow no side-effect tools in analysts.                                             |
| LLM cost        | Deterministic budget guard (per-hour and per-day caps). Cheaper models for triage and analysts, stronger model for panel/Axel. Cache repeated context. Event-driven triggers (new bar close, news event, filing). |
| Reliability     | Idempotent client order IDs, retry with backoff, order state machine, periodic broker reconciliation, data-freshness checks, market calendar awareness.                                                           |

| Area          | Requirement                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Observability | Heartbeats from each process, alert on silence, structured logs, per-call LLM traces, immutable decision log.                                                   |
| Hosting       | Small VPS with automatic restarts and daily DB backups. Test a restore. Keep the watchdog on a separate host or at least a separate service account if you can. |
| Records       | Keep trade records and fills exportable for accounting and tax purposes.                                                                                        |

## 10. Testing strategy

- **Unit and property tests** for the risk package (highest priority; the safety claim depends on it).
- **Replay tests**: feed recorded market days through the whole pipeline with a fake clock and a stubbed LLM; outputs must be deterministic.
- **Contract tests** for broker adapters against the paper API and a recorded-response fake.
- **Chaos drills**: kill processes, drop the network, feed stale data, return malformed LLM output; the system must fail closed (no new orders).
- **Adversarial prompt tests**: injection strings in fake emails and headlines must not change proposals.

## 11. Open decisions (from the PRD's "To be agreed" list and this review)

| Decision                 | Suggested default                                                                                                              | Decide by     |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------|
| Stocks/options execution | Alpaca (paper first). Confirm live availability for your residence before Phase 6.                                             | Phase 1       |
| Crypto execution         | Kraken or Binance, chosen by availability and API quality for your country.                                                    | Phase 7       |
| Prediction venue         | Kalshi if permitted; Polymarket/sports only after legal review.                                                                | Phase 9       |
| Hosting                  | Single VPS + Docker Compose. Move to managed services only if needed.                                                          | Phase 0       |
| Dashboard                | Grafana first; custom UI later if justified.                                                                                   | Phase 4       |
| Orchestration            | LangGraph inside the agent service only; trading core is plain Python services.                                                | Phase 3       |
| ML / RL (PyTorch, FinRL) | Defer. Start with rules and gradient-boosted models on engineered features; add RL only if a baseline is beaten and validated. | After Phase 6 |
| Numeric thresholds       | Fix section 8 values in a config file, committed before first results.                                                         | Phase 2       |
| Live approval policy     | Every trade needs approval for first N weeks; relax by explicit decision.                                                      | Phase 6       |

## 12. Top risks

| Risk                                       | Mitigation                                                                                             |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Overfitting / leakage produces fake edge   | Strict validation gate, untouched OOS block, multiple-testing adjustment, LLM contamination awareness. |
| Scope creep across four sections           | Vertical slice; phase exit criteria; later sections reuse the core.                                    |
| Runaway LLM cost                           | Budget guard, event-driven triggers, model tiering.                                                    |
| Silent failure (stale data, drifted state) | Reconciliation, heartbeats, fail-closed defaults.                                                      |
| LLM-driven bad trade                       | LLM cannot reach execution; deterministic veto; per-trade risk cap; human approval in live mode.       |
| Legal / access issues on venues            | Confirm per venue before building; paper trading first.                                                |

This plan describes software engineering structure only. It is not financial or legal advice, and automated trading can lose money even when validated.